

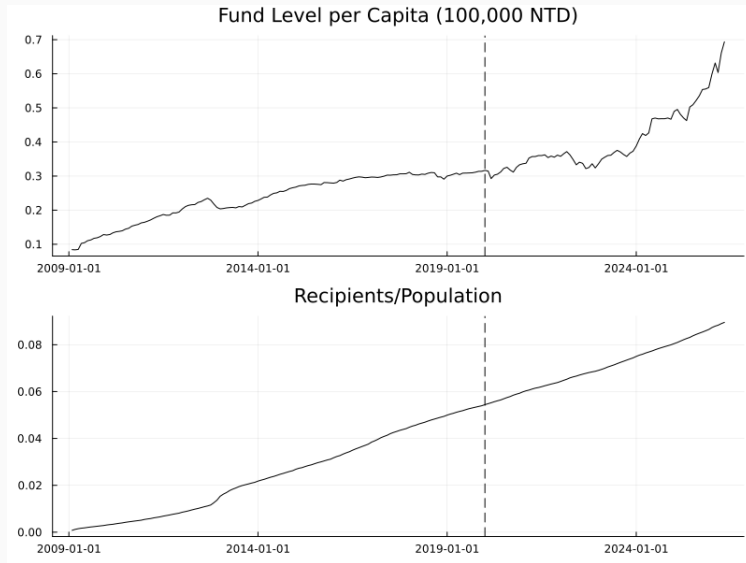
Pension Run

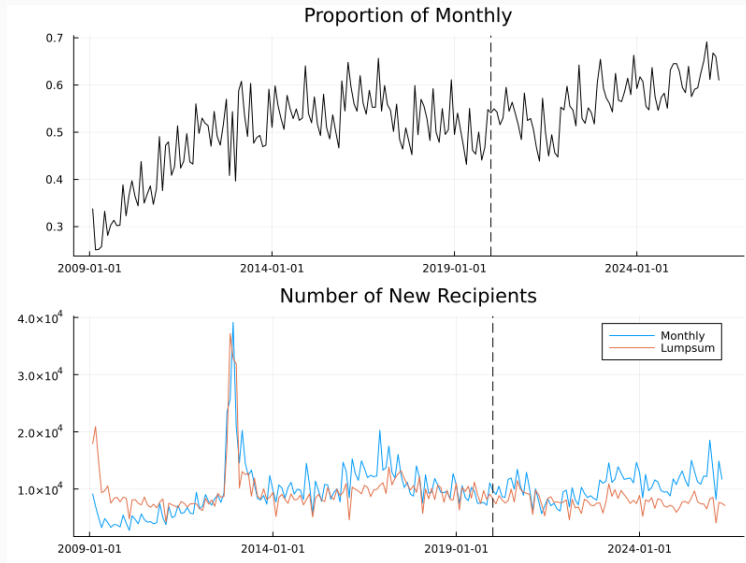
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- How does the potential reform of the pension system affect the retirement decisions?
 - Retire later to accumulate more wealth.
 - Retire earlier to withdraw more pension.
 - Reform Uncertainty.
 - Run on the pension system.
- Related literature:
 - **Pension reform:** Bi and Zubairy (2023); Daminato and Padula (2024); Delalibera, Ferreira, and Machado Parente (2026); French and Jones (2011); Kitao (2018).
 - **Bank run:** He and Xiong (2012); ?.

Background

- Since 2009, one with insurance records before 2009 can choose between lumpsum and monthly scheme.
- For a recipient with average monthly insured salary NTD 45,800 with 30 years of working retired at standard retirement age and lives 12 years after retirement.
 - Lumpsum: 2,061,000 NTD.
 - Monthly: 2,724,504 NTD. (2% annual rate)
- Reform information shocks:
 - Reform committee established in 2012.
 - Reform on Civil Servant Retirement Pension in 2017.
 - External finance from the government since 2020.





Model

Model Overview

- Continuous time and an infinite horizon.
- A continuum of workers chooses once, at retirement age T , between a lump-sum benefit ℓ and a monthly pension p .
- The pension fund earns a stochastic return and receives payroll tax revenue.
- Benefits stop when the fund balance reaches zero.
- Retirees' choices affect fund outflows and, consequently, the value of the monthly pension.

Population

Let $g(t, s)$ denote the population density at age s :

$$\frac{\partial g}{\partial t} = - \underbrace{\frac{\partial g}{\partial s}}_{\text{aging}} - \underbrace{m(s)g}_{\text{mortality}} + \underbrace{\int b(s)g(t, s) ds}_{\text{birth}} \delta_0.$$

Assume $b(s) = b$ and $m(s) = m\mathbf{1}\{s \geq T_m\}$. In a steady-age distribution, N_t grows at rate n and

$$\hat{g}(s) = \begin{cases} be^{-ns}, & s < T_m, \\ be^{-ns}e^{-m(s-T_m)}, & s \geq T_m, \end{cases}$$

where

$$1 = b \frac{1 - e^{-nT_m}}{n} + be^{-nT_m} \frac{1}{n + m}.$$

Pension System

Let q_t be the share of new retirees choosing the monthly pension and Q_t the stock of monthly-benefit recipients. For $T \geq T_m$,

$$dQ_t = \left[\underbrace{q_t \hat{g}(T) N_t}_{\text{new retirees}} - \underbrace{m Q_t}_{\text{mortality}} \right] dt.$$

The pension fund balance evolves according to

$$dB_t = \left[\underbrace{r B_t}_{\text{interest}} + \underbrace{\tau y N_t \int_{T_w}^T \hat{g}(s) ds}_{\text{pension tax}} - \underbrace{(1 - q_t) \ell N_t \hat{g}(T)}_{\text{lump sum}} - \underbrace{p Q_t}_{\text{monthly}} \right] dt + \sigma B_t dZ_t.$$

The system is bankrupt when B_t first reaches zero.

Detrended State Dynamics

Define fund balance and recipients per capita by $\hat{B}_t = B_t/N_t$ and $\hat{Q}_t = Q_t/N_t$. Then

$$d\hat{B}_t = \left[\underbrace{(r - n)\hat{B}_t}_{\text{net return}} + \underbrace{\tau y \int_{T_w}^T \hat{g}(s) ds}_{\text{pension tax}} - \underbrace{(1 - q_t)\hat{g}(T)\ell}_{\text{lump sum}} - \underbrace{p\hat{Q}_t}_{\text{monthly}} \right] dt + \sigma \hat{B}_t dZ_t,$$
$$d\hat{Q}_t = \left[\underbrace{q_t \hat{g}(T)}_{\text{new retirees}} - \underbrace{(m + n)\hat{Q}_t}_{\text{mortality}} \right] dt.$$

- More lump-sum claims reduce the fund immediately.
- More monthly claims raise future liabilities through $\hat{Q}_t$.

Individual Pension Choice

A retiring individual chooses $d \in \{\ell, p\}$ with utility

$$u_{it}(d) = \begin{cases} \alpha + \ell + \beta \epsilon_i(\ell), & d = \ell, \\ V(\hat{B}_t, \hat{Q}_t) + \beta \epsilon_i(p), & d = p. \end{cases}$$

The value of monthly benefits is

$$V(\hat{B}, \hat{Q}) = \mathbb{E} \left[\int_0^{T_{\hat{B}}} e^{-(\rho+m)s} p \, ds \mid \hat{B}, \hat{Q} \right].$$

With i.i.d. standard Gumbel ϵ_i , the monthly-benefit recipient share is

$$q(\hat{B}, \hat{Q}) = \frac{1}{1 + \exp \left(\frac{\alpha + \ell - V(\hat{B}, \hat{Q})}{\beta} \right)}.$$

Equilibrium and Main Mechanisms

In a Markov equilibrium:

1. State dynamics are consistent with retirees' choice q .
2. The value V is consistent with the induced bankruptcy time.
3. Individual choices aggregate to $q(\hat{B}, \hat{Q})$.

Two Externalities

1. Low $\hat{Q} \rightarrow$ far bankruptcy time $\rightarrow$ monthly dominates $\rightarrow$ add stress on the pension fund.
2. High $\hat{Q} \rightarrow$ near bankruptcy time $\rightarrow$ lumpsum dominates $\rightarrow$ accelerate bankruptcy.

Value Function and Numerical Solution

The monthly-pension value solves the nonlinear HJB equation

$$(\rho + m)V = p + \mathcal{A}V, \quad V(0, \hat{Q}) = 0,$$

where

$$\mathcal{A}f = \mu_B \frac{\partial f}{\partial \hat{B}} + \mu_Q \frac{\partial f}{\partial \hat{Q}} + \frac{1}{2} \sigma^2 \hat{B}^2 \frac{\partial^2 f}{\partial \hat{B}^2}.$$

- Nonlinearity arises because μ_B and μ_Q depend on q , which depends on V .
- At a sufficiently large upper boundary, impose $\partial V / \partial \hat{B} \approx 0$.
- Solve by finite differences and successive iteration, equivalent to semi-implicit scheme with $dt = \infty$.

Results

Calibration

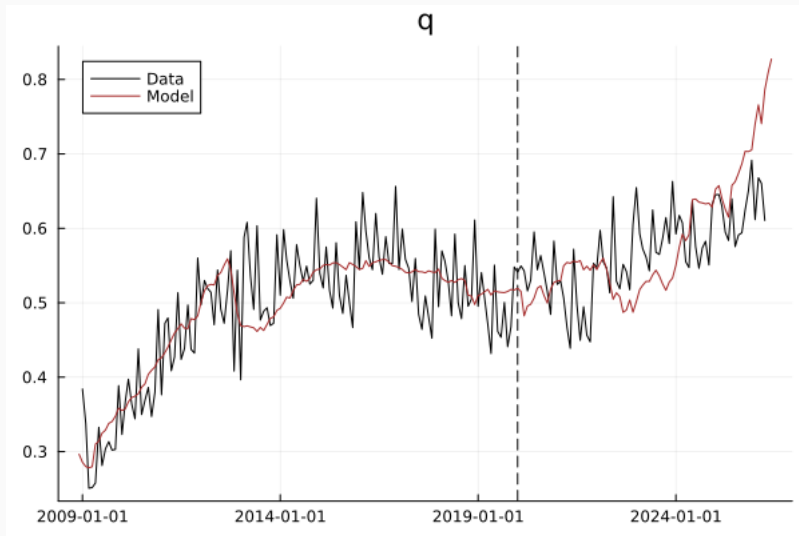
Parameter	Description	Value	Target
<i>Panel A: External Calibration</i>			
b	Birth rate	0.00462	Total fertility rate
m	Mortality rate	0.058	Median lifespan
T_w	Working age	20	
T_r	Retirement age	65	Standard retirement age
T_m	Age of mortality exposure	65	
ρ	Subjective discount rate	0.02	Interest rate
τ	Pension tax	0.115	Pension tax
y	Income	5.496	Highest monthly insured salary
ℓ	Lump-sum pension	20.61	Transfer based on 30 years of working
p	Monthly pension	2.556	Pension based on 30 years of working

Note: The monetary unit in the model is 100,000 NTD.

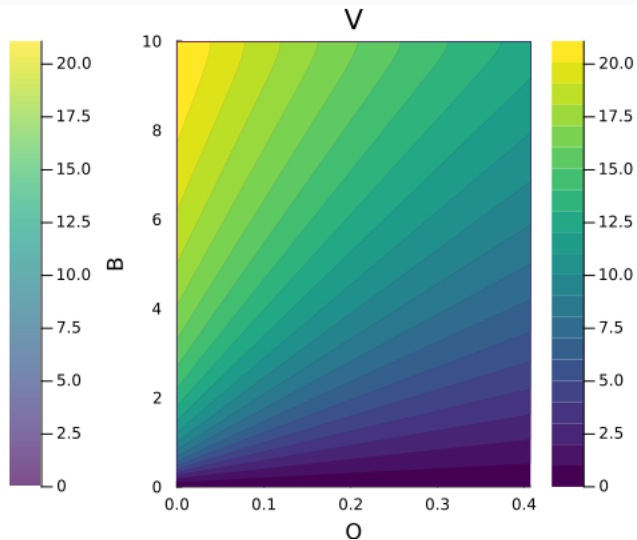
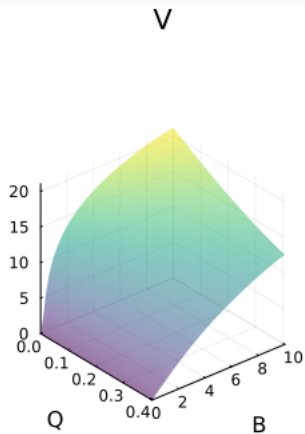
Parameter	Description	Value	Target
<i>Panel B: Internal Calibration</i>			
r	Pension fund return rate	-0.1293	Mean fund-return residual = 0
σ	Pension fund volatility	0.0866	Fund-return residual SD = σ
α	Preference for lump-sum scheme	-18.63	$E \left[(q - q(\hat{B}_t, \hat{Q}_t, \theta))^2 \right]$
β	Sensitivity to default risk	1.221	$E \left[(q - q(\hat{B}_t, \hat{Q}_t, \theta))^2 \right]$

Note: The monetary unit in the model is 100,000 NTD.

Validation



Estimated V



- Experiments
 - No external finance since 2020.
 - Raising pension tax rate.
 - No monthly option ($\alpha \rightarrow \infty$).

- Relax the mortality assumption.
- Belief process.
- Endogenous retirement.
- Regime switch in fund return.